2 Fig. 2.1 shows a space vehicle which consists of a capsule and a nose cone.

The space vehicle is moving at a velocity of 7800 m / s. The mass of the space vehicle is 840 kg.

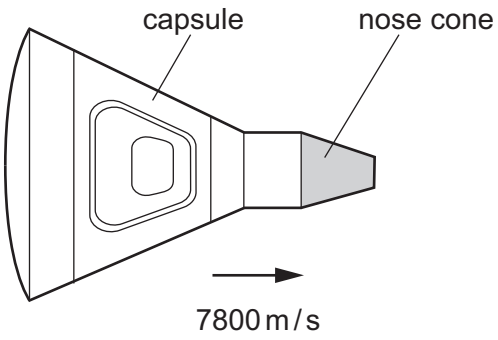

Fig. 2.1

(a) Show that the momentum of the space vehicle is approximately 6.55×10^6 kg m / s.

[1]

(b) The capsule ejects the nose cone, as shown in Fig. 2.2.

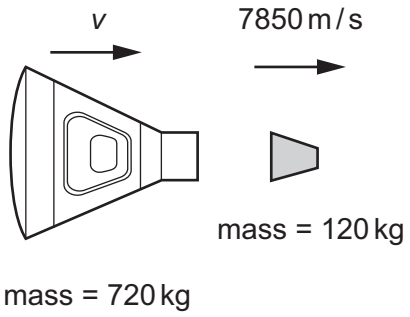

Fig. 2.2 (not to scale)

Determine the velocity v of the capsule after the nose cone is ejected. Give your answer to 3 significant figures.

velocity v of the capsule = ... [3]

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(c) A different space capsule returns to Earth.
Fig. 2.3 shows this capsule just before it lands in the sea.
The capsule travels at terminal velocity.

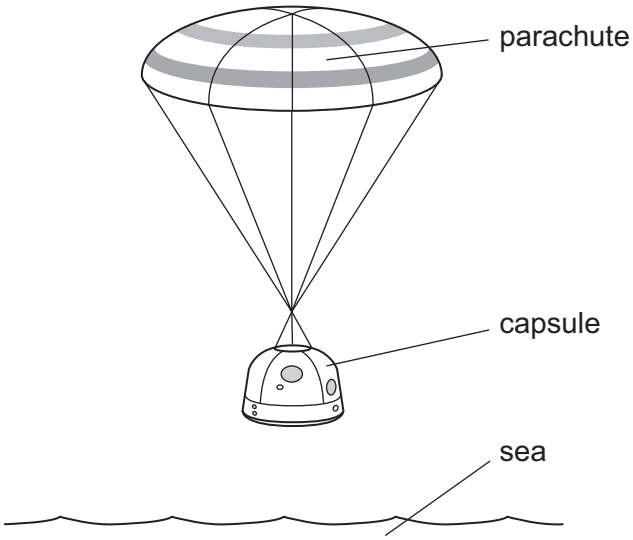

Fig. 2.3

The upward vertical force acting on the capsule is 120 kN.

Calculate the mass of the capsule.

mass of the capsule = ... [2]

[Total: 6]

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